

# Ideation Phase

## Brainstorm & Idea Prioritization

|               |                                                     |
|---------------|-----------------------------------------------------|
| Date          | 18 June 2026                                        |
| Student Name  | JASTI RAGHAVENDRA<br>(jsvpsairaghavendra@gmail.com) |
| Team ID       |                                                     |
| Project Name  | ShopEz                                              |
| Maximum Marks | 4 Marks                                             |

### Brainstorm & Idea Prioritization Template:

A structured brainstorming approach is utilized here to refine the creative process, allowing for the precise definition of system specifications and technical frameworks. This stage emphasizes the selection of high-value capabilities for the ShopEz ecosystem while validating their technical viability within the MERN architecture.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

### Step-1: Selection of the Problem Statement

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**Contextual Analysis:** In today's digital marketplace, shoppers expect a cross-platform experience that is both instantaneous and bulletproof. However, many current platforms struggle with outdated administrative interfaces or sluggish delivery updates, often due to fragmented technology stacks.

**Defined Problem Statement:** "What is the most effective way to architect a full-stack MERN application that ensures a seamless, high-performance shopping journey for users while providing a robust, data-driven administrative control center?"

## Step-2: Brainstorm, Idea Listing and Grouping

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The following categorization details the primary conceptual pillars established during the individual ideation session:

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>User Interface (Frontend)</u></b>                                                                                                                                                                                                                 | <b><u>Backend &amp; Logic</u></b>                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"><li>• Dynamic product showcase built with React</li><li>• Device-agnostic CSS for all demographics</li><li>• Intuitive search and filtering capabilities</li><li>• State-driven cart management using Redux</li></ul> | <ul style="list-style-type: none"><li>• Secure user verification via JWT protocols</li><li>• Highly scalable Express.js API framework</li><li>• Comprehensive purchase tracking system</li><li>• Synchronized inventory adjustment logic</li></ul> |
| <b><u>Database (MongoDB)</u></b>                                                                                                                                                                                                                        | <b><u>Admin Features</u></b>                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"><li>• Optimized data models for users and items</li><li>• Enhanced query speed through index optimization</li><li>• Distributed media storage via Cloudinary</li></ul>                                                | <ul style="list-style-type: none"><li>• Role-based access for management panels</li><li>• Full operational control over product inventory</li><li>• Automated workflows for order completion</li></ul>                                             |

### Step-3: Idea Prioritization

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The matrix below evaluates key development modules based on their functional necessity versus implementation effort to streamline the MVP development cycle.

| Feature Category       | Importance<br>(High/Med) | Feasibility<br>(High/Med) | Priority Level      |
|------------------------|--------------------------|---------------------------|---------------------|
| Core MERN Integration  | High                     | High                      | Critical (Phase 1)  |
| Security & Permissions | High                     | High                      | Critical (Phase 1)  |
| Management Interface   | High                     | Medium                    | High (Phase 2)      |
| Refined Search Logic   | Medium                   | High                      | Standard (Phase 2)  |
| Dynamic Order Updates  | Medium                   | Medium                    | Secondary (Phase 3) |